AI-Based Rehabilitation Exercise Monitoring using Webcam

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Motivation & Problem Statement

Physical rehabilitation exercises are essential for recovery after injuries, surgeries, or neurological disorders. Many patients perform exercises incorrectly when practicing at home without supervision, which can reduce recovery effectiveness or lead to further injury. Traditional physiotherapy requires frequent in-person monitoring, which is often expensive, inaccessible, or time-consuming. This project proposes an AI-based rehabilitation exercise monitoring system that uses a standard webcam to analyze a patient's body posture and movement in real time and provide feedback on whether exercises are being performed correctly.

Proposed Approach

The system will use computer vision techniques to estimate human body keypoints from webcam input and compare detected motion against expected movement patterns, focusing on exercises such as arm raises, squats, and knee rehabilitation movements. The proposed pipeline includes: (1) human pose estimation from live webcam video, (2) exercise classification and motion tracking, (3) detection of incorrect posture or incomplete movement, and (4) real-time feedback to the user. AI techniques will include supervised learning for exercise classification, pose landmark analysis for posture evaluation, and sequence-based classification (e.g., an LSTM) to capture motion dynamics across frames, all implemented from scratch. Depending on progress, the project may also include a scoring system that evaluates exercise quality or repetition accuracy.

Existing Approaches

Existing systems commonly rely on pose estimation frameworks to extract skeletal keypoints from video. MediaPipe Pose (Google) is a lightweight real-time framework extracting 33 body landmarks, well-suited for webcam-based pipelines. OpenPose (CMU) is a widely-used multi-person pose estimation library and provides a strong benchmark. Recent literature has also explored machine learning models for physiotherapy assessment and posture correction [1]. Commercial systems exist for fitness and exercise monitoring but typically require specialized hardware or are not designed for rehabilitation settings. This project targets rehabilitation specifically using only a webcam and AI methods built from scratch.

References

[1] Sievert, T. et al. *Artificial Intelligence and Machine Learning in Rehabilitation.* npj Digital Medicine, 2020. nature.com/articles/s41746-020-0250-6